

Homework 6 - EMT

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8.22)

$$\begin{aligned}\bar{M} &= \chi_m \bar{H} & \therefore \bar{H} &= \frac{\bar{B}}{\mu_0 \mu_r} \\ \bar{M} &= \chi_m \frac{\bar{B}}{\mu_0 \mu_r}\end{aligned}$$

$$\therefore \mu_r = 1 + \chi_m$$

$$\bar{M} = \frac{\chi_m \bar{B}}{\mu_0 (1 + \chi_m)}$$

$$\boxed{\bar{M} = \frac{\chi_m \bar{B}}{\mu_0 (1 + \chi_m)}}$$

8.25) $\mu = 4.5\mu_0$, $\bar{B} = 4y \text{ a}_z \text{ mWb/m}^3$
calculate:

(a) χ_m (b) $\bar{H}$ (c) $\bar{M}$ (d) $\bar{J}$

$$(a) \mu_r = 1 + \chi_m$$

$$\chi_m = \mu_r - 1$$

$$\chi_m = 4.5 - 1$$

$$\boxed{\chi_m = 3.5}$$

$$\therefore \mu_0 \mu_r = \mu = 4.5\mu_0$$

$$\mu_r = 4.5$$

$$(b) \frac{\bar{H} = \bar{B}}{\mu} = \frac{4y \cdot a_z \times 10^{-3}}{4.5\mu_0} \therefore \mu_0 = 4\pi \times 10^{-7}$$

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$$\vec{H} = \frac{4 \times 10^{-3}}{4\pi \times 10^{-7} \times 4.5} y \cdot a_z \approx \boxed{707.3 y \cdot a_z \text{ A/m}}$$

$$\begin{aligned} \text{(c)} \quad \vec{M} &= \chi_m \vec{H} \\ &= (3.5) (707.3 y \cdot a_z) \\ &= 2.476 \times 10^3 y a_z \text{ A/m} \end{aligned}$$

$$\boxed{\vec{M} = 2.476 y a_z \text{ kA/m}}$$

$$\text{(d)} \quad \vec{J}_b = \nabla \times \vec{M} = \begin{vmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & 2.476ky \end{vmatrix}$$

$$= \frac{\partial}{\partial y} (2.476 \times 10^3 y) \cdot a_x$$

$$\boxed{\vec{J}_b = 2.476 \times 10^3 \cdot a_x \text{ A/m}^2}$$

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8.41: Determine the inductance per unit length of a cylindrical conductor of radius 3mm. Assume $\mu = \mu_0$.

Inductance inside the cylindrical conductor is given by: $L = \frac{\lambda}{I}$

$$\lambda_{\text{inside}} = \frac{\mu_0 I l}{8\pi} \quad \text{hence, } L = \frac{\mu_0 l}{8\pi}$$

$$L = \frac{(4\pi \times 10^{-7})}{2 \times 8\pi} l = \boxed{50 \text{ nH}}$$